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Applications of AI to single-cell and spatial transcriptomics: current state-of-the-art and challenges

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Artificial intelligence (AI) has become a common tool for bioinformatics, with hundreds of methods published in recent years. Due to the training data demands of deep-learning algorithms, high-throughput single-cell and spatial transcriptomics is one of the most popular areas for these applications. Here we review how AI is being used for single-cell and spatial transcriptomics analysis, and how these approaches compare to alternative statistical or heuristic-based methods. We explored 10 common analysis tasks: dimensionality reduction, cross-dataset integration, data denoising, data augmentation, deconvolution, cell-cell interactions, transcriptional velocity, transcriptomic-chromatin accessibility integration, and integrating single-cell and spatial transcriptomics modalities. We highlight which algorithms are likely to be useful for discovery researchers, and which are not yet ready for general research use.

KEYWORDS

cell-cell interactions, cross-dataset integration, data denoising, deconvolution, dimensionality reduction, integrating single-cell and spatial transcriptomics modalities, transcriptional velocity

1 Introduction

Artificial intelligence (AI) has revolutionized the analysis of big data across many fields, including biomedical research, and is entering clinical practice, with over 1,000 algorithms and devices approved by the FDA (Health, 2025). While the predominant use of AI in clinical practice is in biomedical image analysis, in research, AI approaches have gained increasing popularity in bioinformatics, and especially single-cell and spatial transcriptomics (Ge et al., 2024; Erfanian et al., 2023; Zahedi et al., 2024; Molho et al., 2024; Ma and Xu, 2022). AI is often used synonymously or as a subtopic of the broader field of machine learning. Machine learning involves a computer or algorithm deriving at least some aspects of a model from observed or “training” data. This includes tasks as simple as estimating the slope and intercept of the best-fit line, or those as complex as labelling MRI images with specific pathological lesions. AI, or deep learning (DL) as we will refer to it, is a specific class of models based on neural networks (NN) with multiple interconnected layers of functions capable of learning complex, non-linear patterns within large-scale datasets.